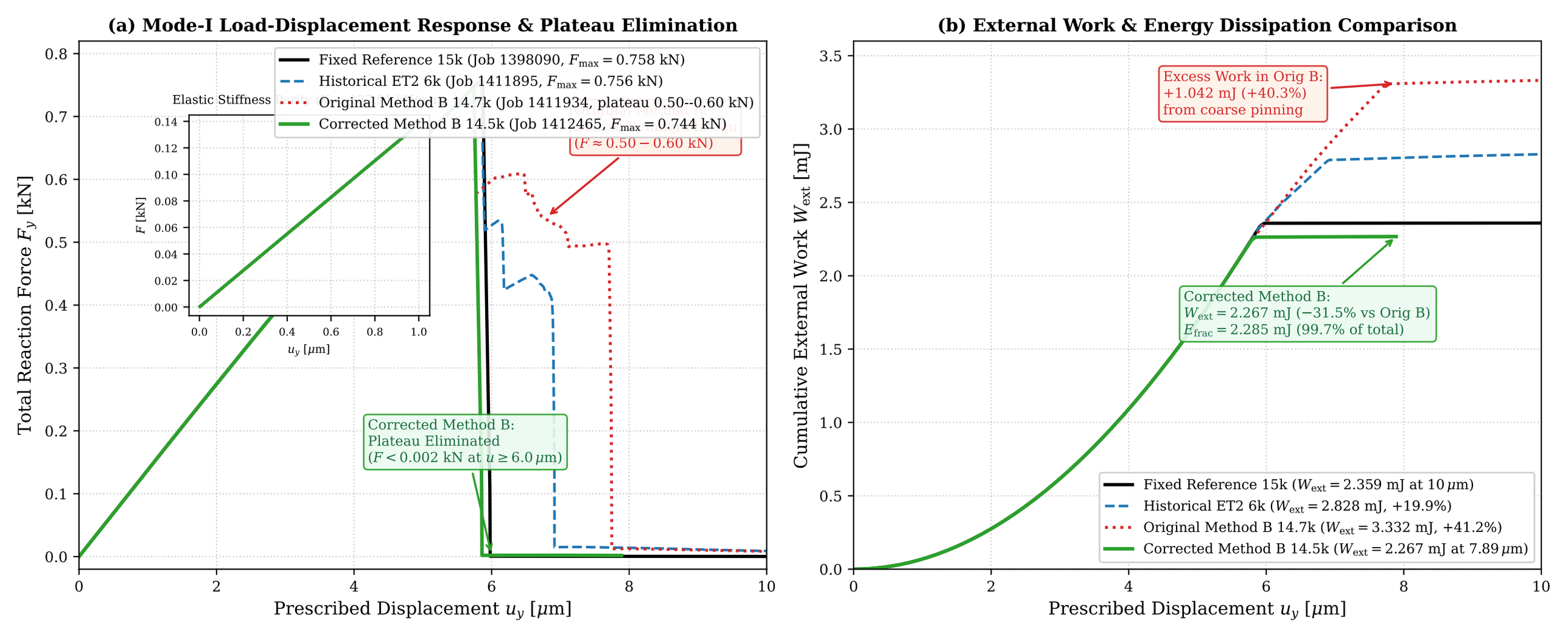
(a) Mode-I Load-Displacement Response & Plateau Elimination
Fixed Reference 15k (Job 1398090, $F_{max} = 0.758$ kN)
Historical ET2 6k (Job 1411895, $F_{max} = 0.756$ kN)
Original Method B 14.7k (Job 1411934, plateau 0.50--0.60 kN)
Corrected Method B 14.5k (Job 1412465, $F_{max} = 0.744$ kN)
Elastic Stiffness
F [kN]
u_y [μm]
($F \approx 0.50 - 0.60$ kN)
Corrected Method B:
Plateau Eliminated
($F < 0.002$ kN at $u \geq 6.0\,\mu$m)
Total Reaction Force F_y [kN]
Prescribed Displacement u_y [μm]
(b) External Work & Energy Dissipation Comparison
Excess Work in Orig B:
+1.042 mJ (+40.3%)
from coarse pinning
Corrected Method B:
$W_{ext} = 2.267$ mJ (−31.5% vs Orig B)
$E_{frac} = 2.285$ mJ (99.7% of total)
Fixed Reference 15k ($W_{ext} = 2.359$ mJ at 10 μm)
Historical ET2 6k ($W_{ext} = 2.828$ mJ, +19.9%)
Original Method B 14.7k ($W_{ext} = 3.332$ mJ, +41.2%)
Corrected Method B 14.5k ($W_{ext} = 2.267$ mJ at 7.89 μm)
Cumulative External Work W_{ext} [mJ]
Prescribed Displacement u_y [μm]